

Portfolio Analytics Dashboard

1. Introduction

This project focuses on portfolio performance and attribution analysis, presented through an interactive Power BI dashboard. It evaluates investment performance using benchmark index data and portfolio data derived from an investment fund's performance.

2. Methodologies

- Data Processing: Used Power Query and Excel for data cleaning and transformation.
- Calculations & Analysis: Implemented DAX formulas for portfolio weight calculations, Brinson attribution analysis, and key rate duration (KRD) analysis for fixed-income exposure.
- Visualization: Designed an interactive Power BI dashboard to present performance, risk, and scenario analyses.

3. Key Components/Output

1. Summary – Provides an overview of portfolio performance.
2. Equity Analysis – Conducts Brinson attribution analysis for equity holdings.
3. Fixed Income Analysis – Evaluates fixed-income asset performance by sector and risk factors.
4. Risk Analysis – Assesses portfolio risks using volatility, tracking error, beta and value at risk (VaR).
5. Scenario Analysis – Examines portfolio performance under different economic conditions.

4. Limitations and Future Improvements

- Data Constraints: The dataset covers only nine months with monthly data points, limiting the accuracy of risk metrics such as volatility and VaR.
- Scope: The analysis is limited to Canadian equity and fixed-income assets. Future improvements include expanding the dataset to cover global markets, incorporating additional asset classes, and extending the time horizon for deeper risk analysis.

Portfolio Summary 2024

Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep
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Select a Date, Return Type, and Securities Type to view performance

MoM

YTD

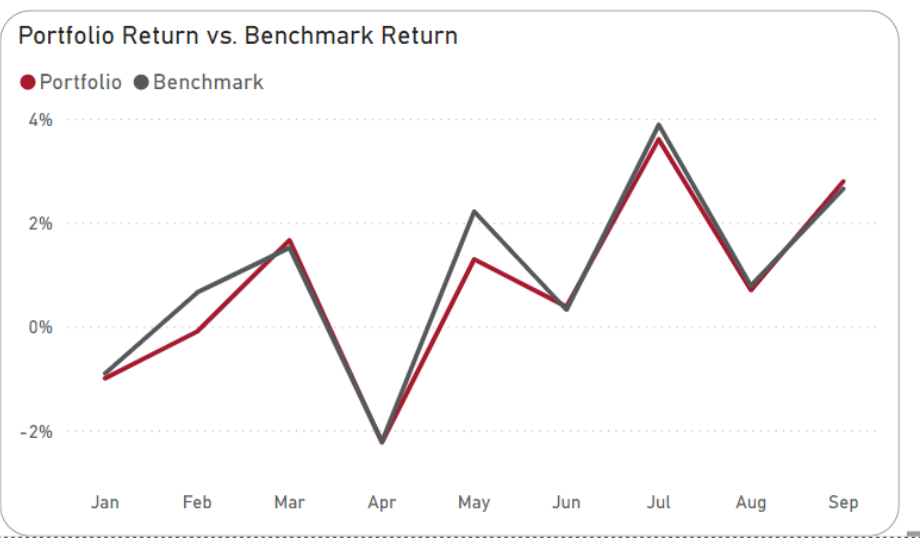
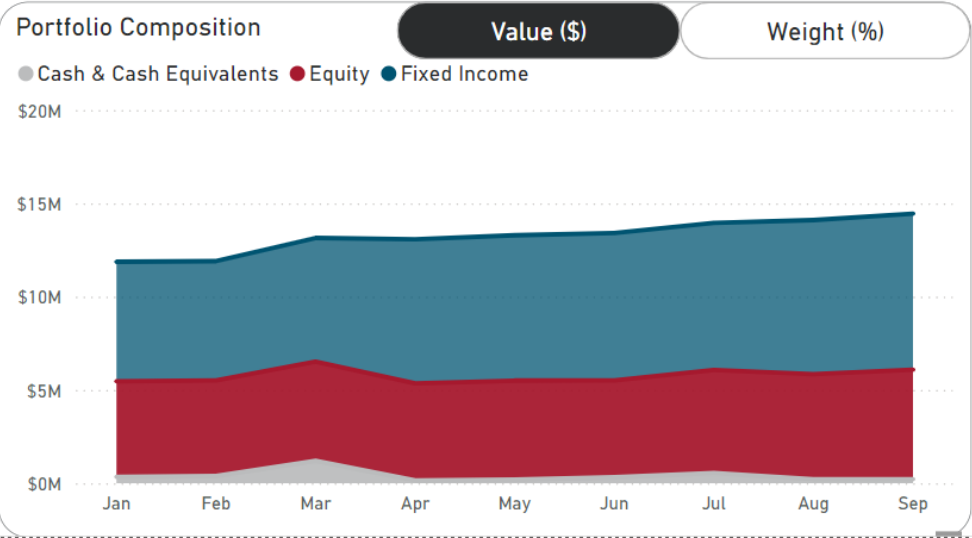
EQUITY

FIXED INCOME SECURITIES



Portfolio Value (Previous Month) \$13,411,668		Portfolio Value (Current) \$13,960,231		Portfolio Value Change \$548,563	
Portfolio Return 3.60%		Benchmark Return 3.88%		Active Return -0.28%	
Portfolio Volatility 1.95%	Tracking Error 0.45%		Information Ratio -0.63		# of Trades 2

Note: Risk metrics are calculated using month-over-month (MoM) data, based on the most recent 6 months, and are not annualized.



Equity

Fixed Income

2024

Jan

Feb

Mar

Apr

May

Jun

Jul

Aug

Sep

MoM Portfolio...
6.27%

MoM Benchmark...
6.38%

MoM Active Return
-0.11%

YTD Portfolio...
9.45%

YTD Benchmark...
10.64%

YTD Active Return
-1.20%

Tracking Error
0.69%

of Holdings
40

Sector/Security Return (%)

Dill up and down to view Return by Sector and Security

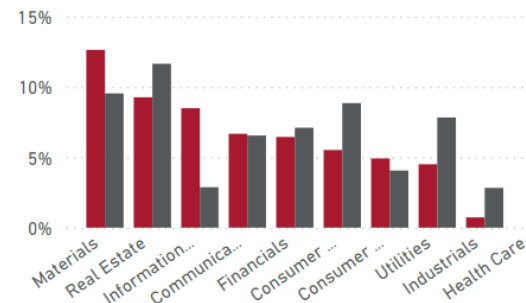
-6.47% 6.29% 19.05%

AEM
BN
WFG
CIX
GIB/A
WPM

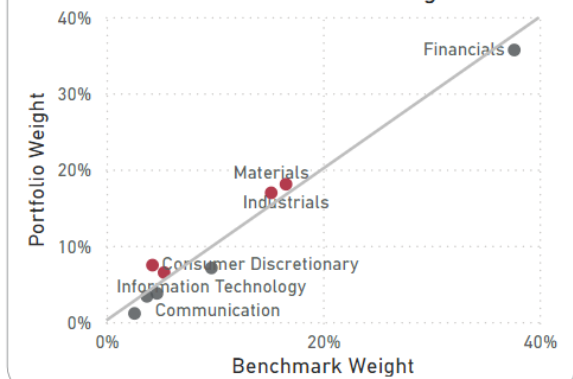


Portfolio Return vs. Benchmark Return by Sector

● Portfolio ● Benchmark

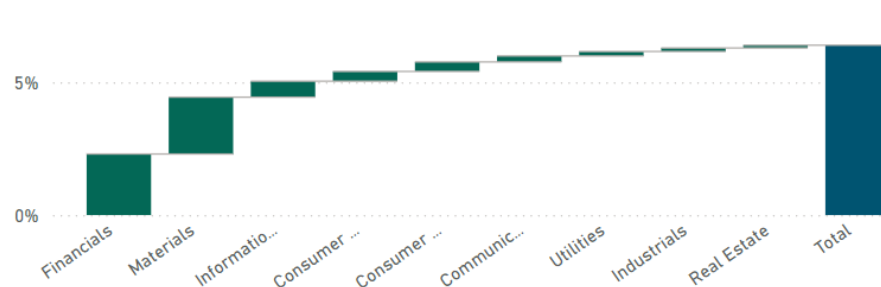


Portfolio vs. Benchmark Sector Weights



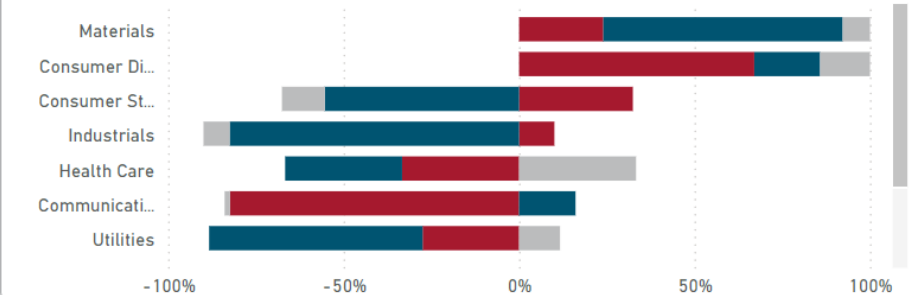
Return Contribution by Sector

● Increase ● Decrease ● Total



Active Return Attribution

● Allocation ● Selection ● Interaction



Equity

Fixed Income

All



Jan

Feb

Mar

Apr

May

Jun

Jul

Aug

Sep



MoM Portfolio Return

1.36%

MoM Benchmark Return

1.09%

MoM Active Return

0.27%

Portfolio Duration

6.89

Benchmark Duration

7.23

Duration Position

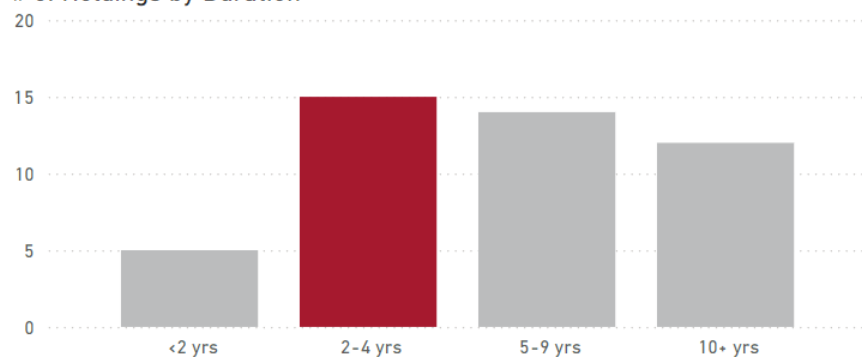
-0.33



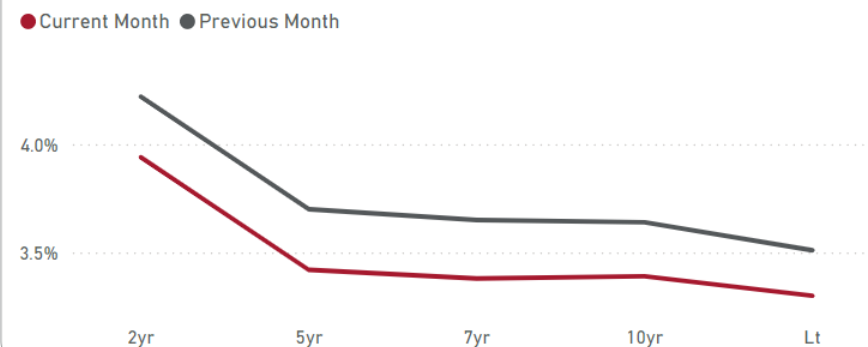
Sector

Factor

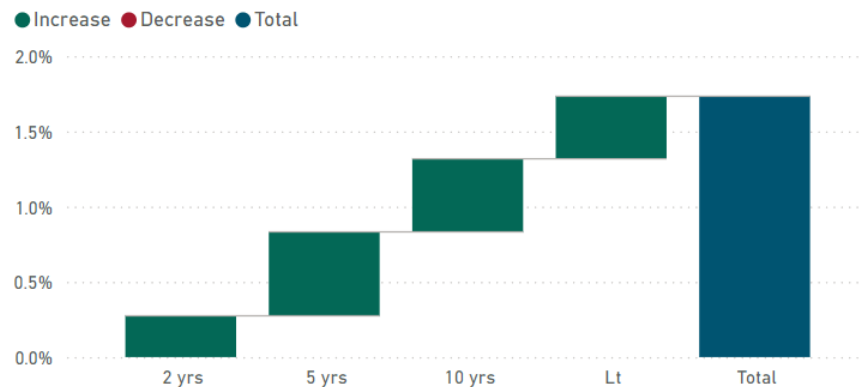
of Holdings by Duration



Term Structure of Interest Rate

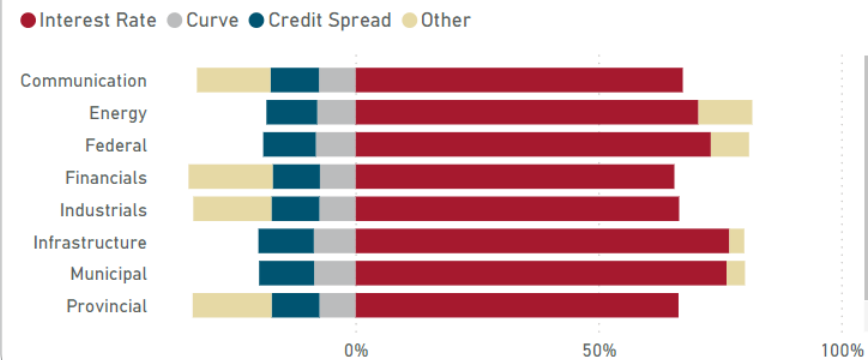


Key Rate Duration Analysis



Return Attribution by Factor

Dill up and down to view attribution by Sector Group and Sector



Risk Analysis

2024

Jan

Feb

Mar

Apr

May

Jun

Jul

Aug

Sep

EQUITY

FIXED INCOME SECURITIES

Portfolio Return

3.60%

Benchmark

3.88%

Portfolio Volatility

1.95%

Benchmark

1.78%

Portfolio Sharpe Ratio

-0.09

Benchmark

0.06

Active Return

-0.28%

Tracking Error

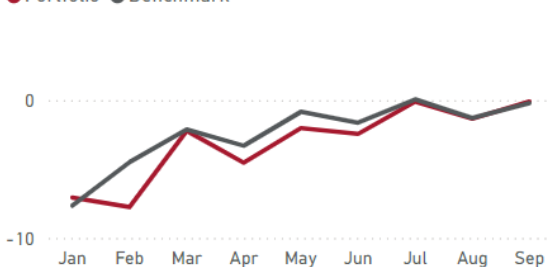
0.45%

Information Ratio

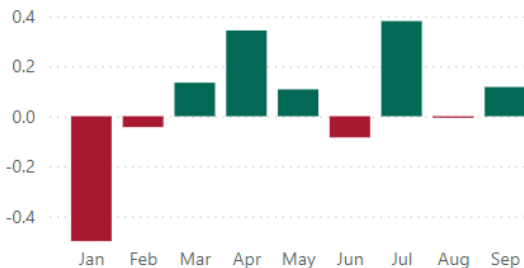
-0.63

Portfolio vs. Benchmark Sharpe Ratio

● Portfolio ● Benchmark

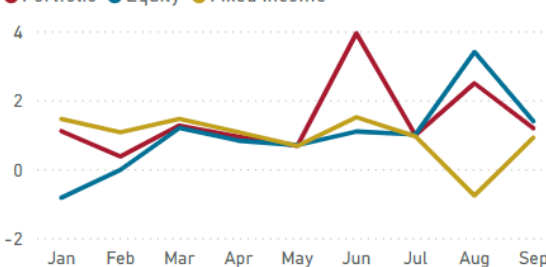


Correlation Between Equity & Fixed Income

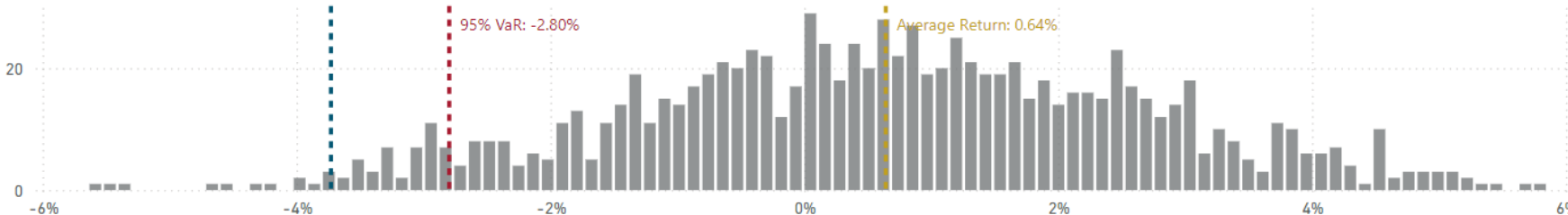


Beta Over Time

● Portfolio ● Equity ● Fixed Income



Portfolio VaR



Scenario Analysis

Results



Scenario	Equity Portoflio	Fixed Income Portfolio	Total Impact
Base Case	2.95%	0.10%	1.25%
Inflation	5.31%	-3.61%	0.11%
Recession	-4.36%	2.24%	-0.50%
Stagflation	0.66%	-0.88%	-0.23%

Assumptions

Scenario	2yr Rate	5yr Rate	10yr Rate	30yr Rate	Growth Stock	Cyclical Stock	Defensive Stock
Base Case	-0.10%	-0.05%	0.00%	0.05%	5.00%	3.00%	1.00%
Inflation	0.70%	0.60%	0.50%	0.40%	12.00%	5.00%	3.00%
Recession	-0.50%	-0.40%	-0.30%	-0.20%	-10.00%	-5.00%	5.00%
Stagflation	0.30%	0.20%	0.10%	0.00%	-2.00%	0.00%	8.00%